

UMC 0.11um Logic and Mixed-Mode AI Advanced Enhancement MIM Capacitor SPICE Model Document

**Version 1.0 Phase 1
2016/05/23**

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1 Introduction

1.1 Revision History

Table 1-1 Revision History

Version	Date	Author	Remark
0.1_p1	2010/01 /08	Jian Zhi Huang	1. Copy model from l110eal_mimcap_vt21.lib in the G-05-LOGIC/MIXED_MODE11-MMC/AL/L110E-SPICE 2. DSM version definition from 2009/11/30.
0.2_p1	2010/07/22	Kelly Pao	1. Based on new silicon data. 2. DSM version definition from 2009/11/30
0.3_p1	2010/12/08	Hung Wei Liao	1. Add 2.0fF/um ² MIMCAP model
0.4_p1	2012/09/28	Ji Wei Hsu	1. Modify 2.0fF/um ² MIMCAP model
0.5_P1	2013/08/09	Ting Huei Wang	1. Change 20f MIMCAP Model minimum size from 4.8*4.8(um ²) to 3.4*3.4(um ²)
0.6_p1	2015/04/15	Zung Jie Han	1. Change 20f MIMCAP Model "cf" parameter from 4.96e-11 to 4.95e-11
1.0_p1	2016/05/23	Tzu Kang Hsiung	1. Modify 15f and 10f MIMCAP Model based on new silicon data . 2. Add trend chart of mapping data into the document . 3. Revise the verification figure (with VC & with TC) of 2.0fF MIMCAP .

1.2 General Information

This document serves as a reference to the design of products that will be manufactured by UMC using the following process.

UMC 0.11 um Logic and Mixed-Mode 1P8M Metal Metal Capacitor AI
Advanced Enhancement Process

For more information about this process and technology, please see the electrical design rule listed below.

G-02-LOGIC/MIXED_MODE11-1P8M-MMC/AL/L110AE-EDR

The creation, verification, and usage of the compact model of the following devices are presented in this document.

1.3 Model Usage

The compact model is provided in various formats. It is guaranteed that the simulation results of this model from different simulators show differences less than 0.5% and 1% for DC and AC simulations respectively. It should be noted that the model has been verified on different simulators of the specific versions listed below and the simulation results from other versions of the simulators might be different.

Synopsys HSPICE release 2014.9SP2
Cadence SPECTRE version 13.1.0.066
Mentor Graphics ELDO version 2013.1

To cope with the impact of process variations on circuit performance, worst-case models are provided along with the typical case and they are listed below.

tt: Typical MIM case
ff: Minimum MIM case
ss: Maximum MIM case

Follow these steps to invoke the model file during simulation.

For Synopsys HSPICE

- Copy ModelFileName.mdl and ModelFileName.lib into your simulation directory.
- Add the following statement in the input file to the simulator.
.lib "/ModelFileName.lib" CaseName

, where CaseName could be tt, ff, or ss.

- Define device condition by 'Xxx' statement.

- Follow the example below to call the model:

X1 top bottom mimcaps_20f_mml110e l=10e-6 w=10e-6

For Cadence Spectre

- Copy ModelFileName.mdl.scs and ModelFileName.lib.scs into your simulation directory.

- Add the following statement in the input file to the simulator.

include "./ModelFileName.lib.scs" section=CaseName

, where CaseName could be mimcaps_ttp, mimcaps_max, or mimcaps_min.

- Define device condition by 'Xxx' statement.

- Follow the example below to call the model:

X1 top bottom mimcaps_20f_mml110e l=10e-6 w=10e-6

For Mentor Graphics Eldo

- Copy ModelFileName.mdl.eldo and ModelFileName.lib.eldo into your simulation directory.

- Add the following statement in the input file to the simulator.

.lib "./ModelFileName.lib.eldo" CaseName

, where CaseName could be mimcaps_ttp, mimcaps_max, or mimcaps_min.

- Define device condition by 'Xxx' statement.

- Follow the example below to call the model:

X1 top bottom mimcaps_20f_mml110e l=10e-6 w=10e-6

ModelFileName for MIM Capacitor:

l110ae_mimcap_v101

DeviceName for MIM Capacitor:

mimcaps_10f_mml110e

mimcaps_15f_mml110e

mimcaps_20f_mml110e

This model is characterized as the following geometry and temperature ranges depicted.

Device behavior beyond these ranges may not be well described by the model.

Geometry Range(before 90% shrink):

w: 3.4um ~ 49.2um

l: 3.4um ~49.2um

Temperature Range:

-50 °C ~ +125 °C

2 Model Equation and Parameter

2.1 Metal/Metal Capacitor

$$C(V) = C0 * (1 + VC1 * Vs + VC2 * Vs^2) * (1 + TC1 * (T - 25) + TC2 * (T - 25)^2)$$

$$C0 = Ca * (L * 0.9 * W * 0.9) + 2 * Cf * (L * 0.9 + W * 0.9)$$

Where:

C(V) : Capacitance

C0 : zero bias Capacitance

Ca : Specific Capacitance

Cf : Fringing Capacitance

W : designed MIMCAP width in um

L : designed MIMCAP length in um

Vs : Bias voltage

T : Element Temperature (From -50°C to 125°C)

VC1 : 1st order (Linear) voltage coefficient of capacitor

VC2 : 2nd order (Quadratic) voltage coefficient of capacitor

TC1 : 1st order (Linear) temperature coefficient of capacitor

TC2 : 2nd order (Quadratic) temperature coefficient of capacitor

1.0fF Metal/Metal Capacitance

	Min.	Typ.	Max.	Unit
Capacitance	0.85	1.0	1.15	fF/um ²
Fringing Capacitance	---	0.0463	---	fF/um
MMC TC1(@F=100KHz)	---	11.2	---	ppm/°C
MMC TC2(@F=100KHz)	---	0.009	---	ppm/°C ²
MMC VC1(@F=100KHz)	---	-28.10	---	ppm/V
MMC VC2(@F=100KHz)	---	-2.17	---	ppm/V ²

1.5fF Metal/Metal Capacitance

	Min.	Typ.	Max.	Unit
Capacitance	1.275	1.5	1.725	fF/um ²
Fringing Capacitance	---	0.0398	---	fF/um
MMC TC1(@F=100KHz)	---	15.8	---	ppm/°C
MMC TC2(@F=100KHz)	---	0.0029	---	ppm/°C ²
MMC VC1(@F=100KHz)	---	-6.84	---	ppm/V
MMC VC2(@F=100KHz)	---	5.67	---	ppm/V ²

2.0fF Metal/Metal Capacitance

	Min.	Typ.	Max.	Unit
Capacitance	1.7	2.0	2.3	fF/um ²
Fringing Capacitance	---	0.0495	---	fF/um
MMC TC1(@F=100KHz)	---	18.0	---	ppm/°C
MMC TC2(@F=100KHz)	---	-0.00145	---	ppm/°C ²
MMC VC1(@F=100KHz)	---	-8.97	---	ppm/V
MMC VC2(@F=100KHz)	---	14.2	---	ppm/V ²

3 MIMCAP Model Verification

3.1 Silicon Wafer for Modeling

The information of the silicon wafer used for characterization and model parameter extraction is summarized in the table below.

Table 3-1 Test wafer information for 1.0fF MIMCAP

	MIMCAP
Mask ID	BSW0FA
Lot ID	D7NJ7.2
Wafer Number	2, 3,4

Table 3-2 Test wafer information for 1.5fF MIMCAP

	MIMCAP
Mask ID	BSW0FA
Lot ID	D5RPN.5
Wafer Number	13, 14, 15

Table 3-3 Test wafer information for 2.0fF MIMCAP

	MIMCAP
Mask ID	BSW0FA
Lot ID	D5RPM
Wafer Number	22, 23, 24, 25

3.2 1.0fF MIMCAP Model Verification

MIMCAP 1.0fF Verification with VC

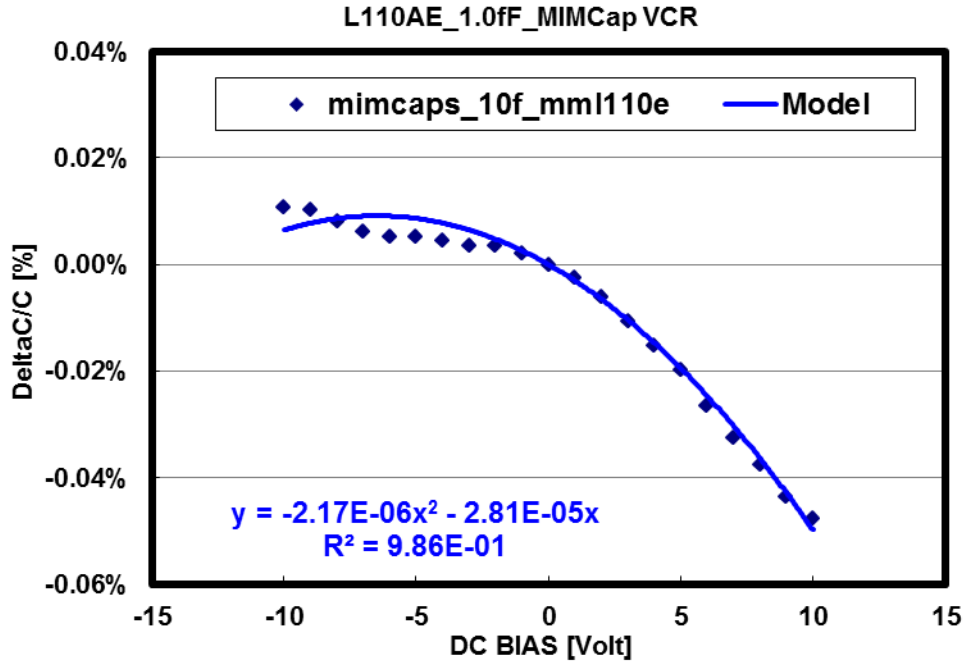


Figure 3-1 1.0 fF/um² MIM Capacitance vs. Voltage curve at 25°C

MIMCAP 1.0fF Verification with TC

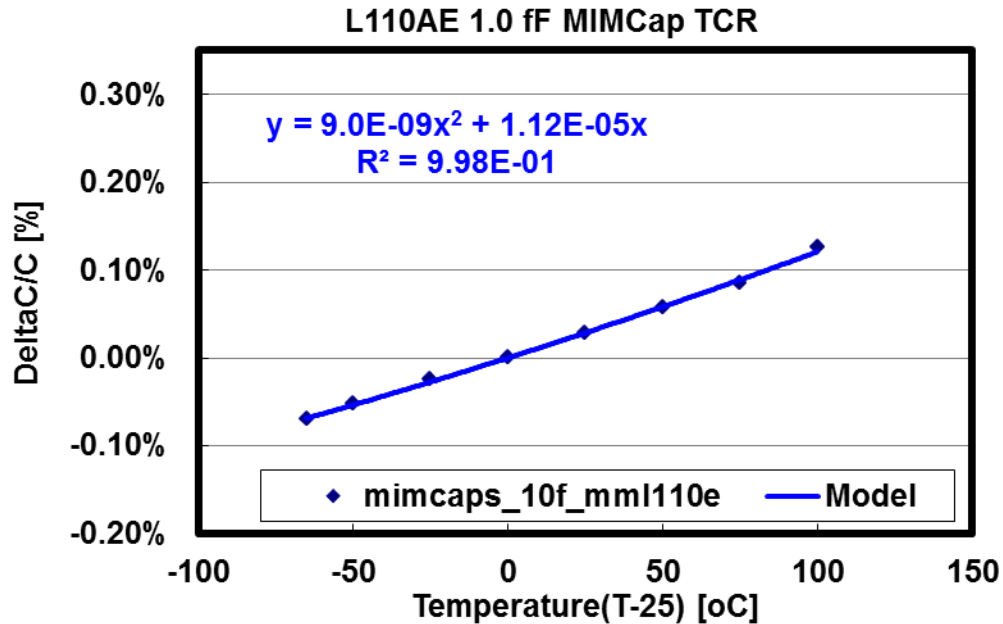


Figure 3-2 1.0 fF/um² MIM Capacitance vs. Temperature curve at V=0V

MIMCAP 1.0fF WAT Trend Chart

Table 3-4 Wafer information for the trend chart of the 1.0fF MIMCAP

	MIMCAP	MIMCAP	MIMCAP
Mask ID	BSW0FA	BSW0FA	BSW0FA
Lot ID	D7NJ7.1	D7NJ7.2	D7NJ7.18
Wafer Number	14,15	2, 3	18, 19

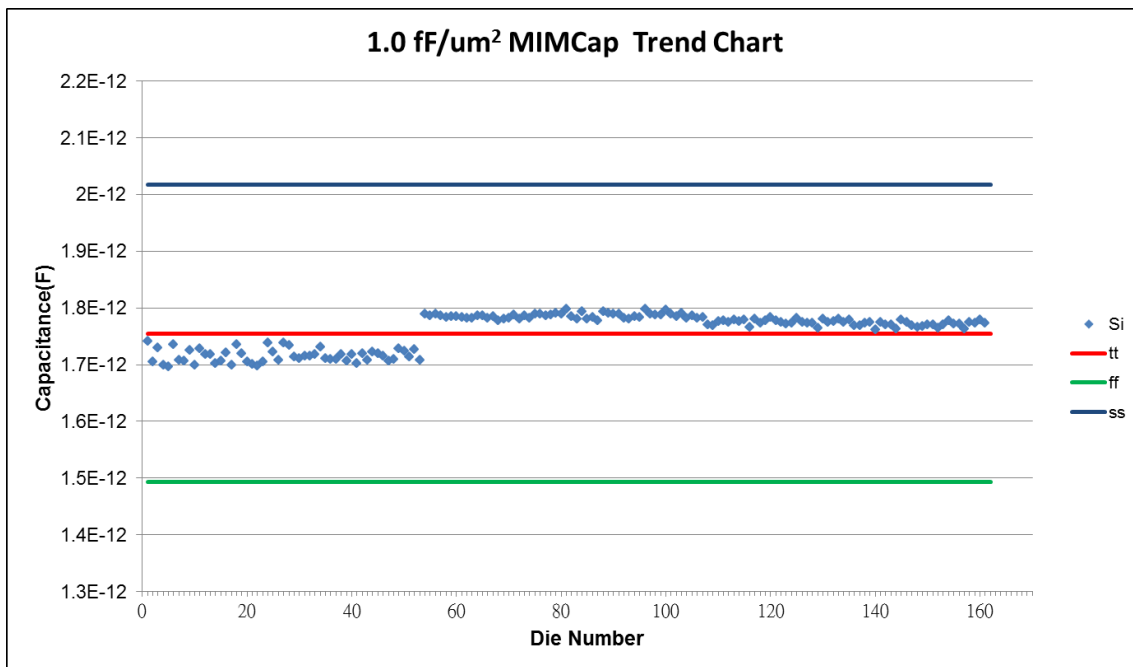


Figure 3-3 1.0 fF/um² MIMCAP_Trend Chart(Before shrink :W=48.6um , L=44.4um)

3.3 1.5fF MIMCAP Model Verification

MIMCAP 1.5fF Verification with VC

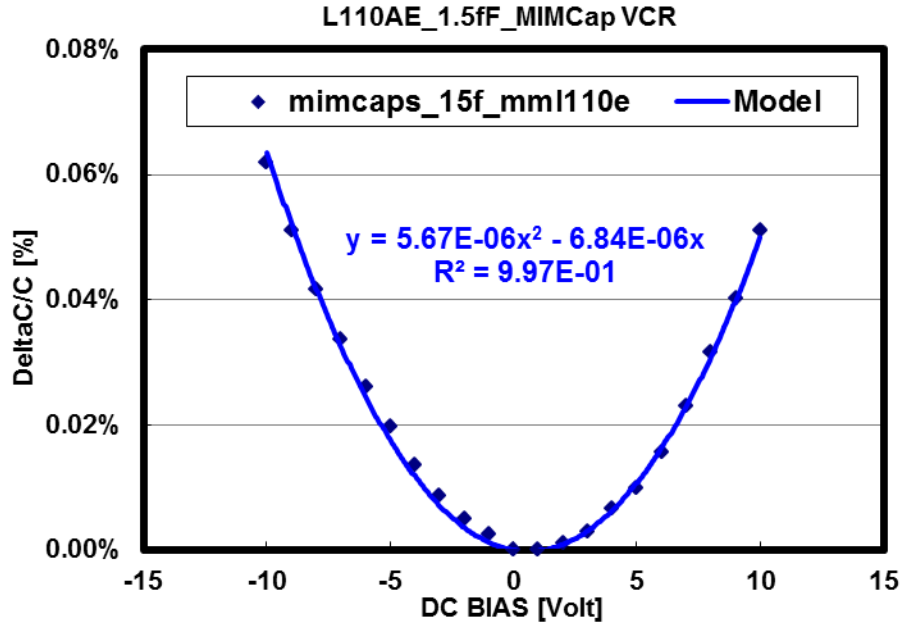


Figure 3-4 1.5 fF/um² MIM Capacitance vs. Voltage curve at 25°C

MIMCAP 1.5fF Verification with TC

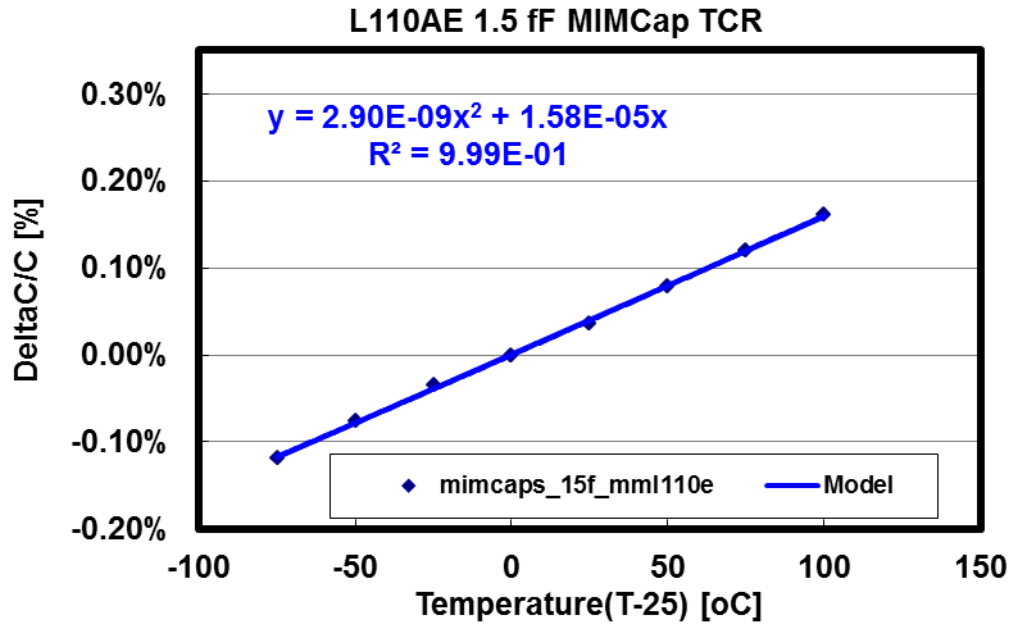


Figure 3-5 1.5 fF/um² MIM Capacitance vs. Temperature curve at V=0V

MIMCAP 1.5fF WAT Trend Chart

Table 3-5 Wafer information for the trend chart of the 1.5fF MIMCAP

	MIMCAP	MIMCAP	MIMCAP
Mask ID	BSW0FA	BSW0FA	BSW0FA
Lot ID	D5RPN.1	D5RPN.5	D5RPN.10
Wafer Number	3, 4	13, 14	22, 23

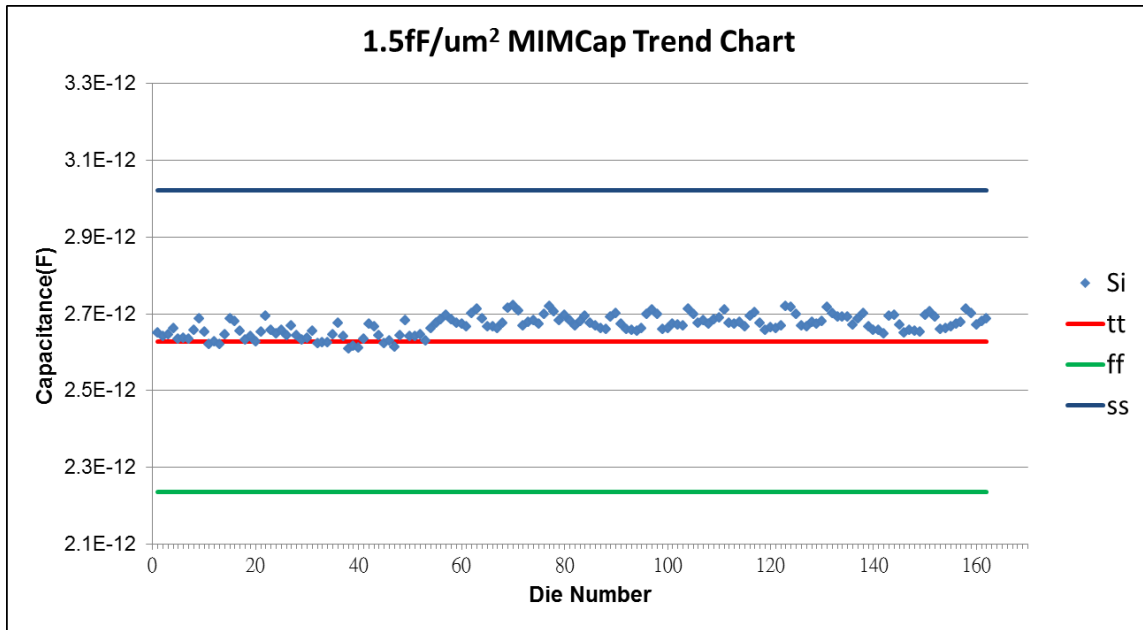


Figure 3-6 1.5 fF/ μm^2 MIMCAP_Trend Chart(Before shrink :W=48.6um , L=44.4um)

3.4 2.0fF MIMCAP Model Verification

MIMCAP 2.0fF Verification with VC

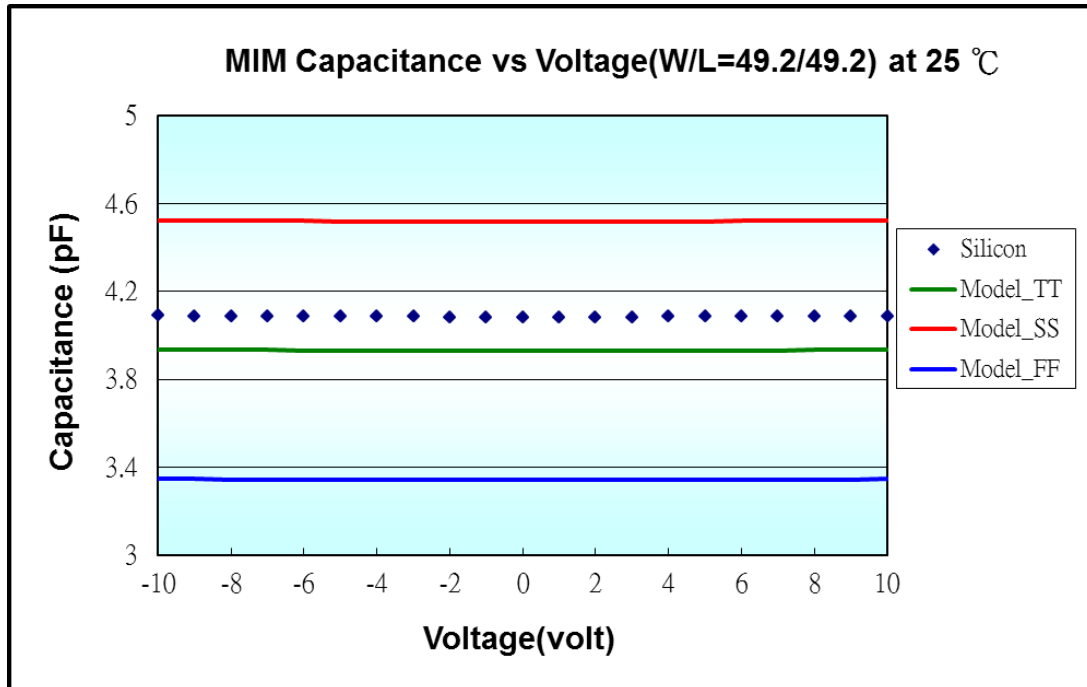


Figure 3-7 2.0 fF/um² MIM Capacitance vs. Voltage curve at 25°C

MIMCAP 2.0fF Verification with TC

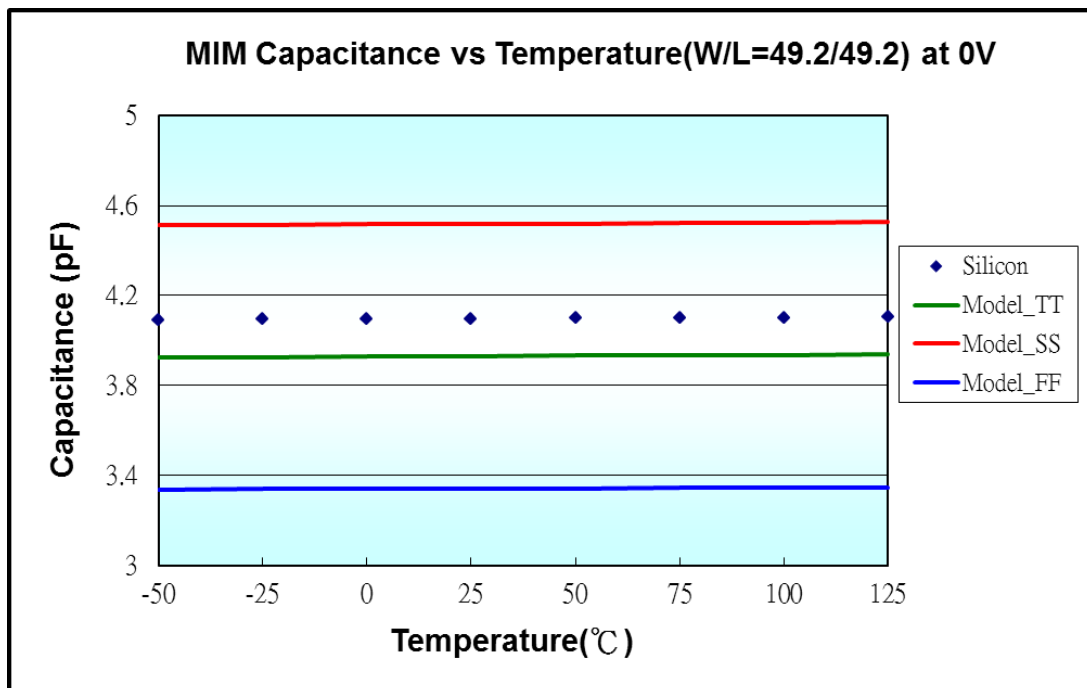


Figure 3-8 2.0 fF/um² MIM Capacitance vs. Temperature curve at V=0V

MIMCAP 2.0fF WAT Trend Chart

Table 3-6 Wafer information for the trend chart of the 2.0fF MIMCAP

	MIMCAP	MIMCAP	MIMCAP
Mask ID	BSW0FA	BSW0FA	BSW0FA
Lot ID	D5RPN.1	D5RPN.5	D5RPM.1
Wafer Number	7,8	17,18	22,23

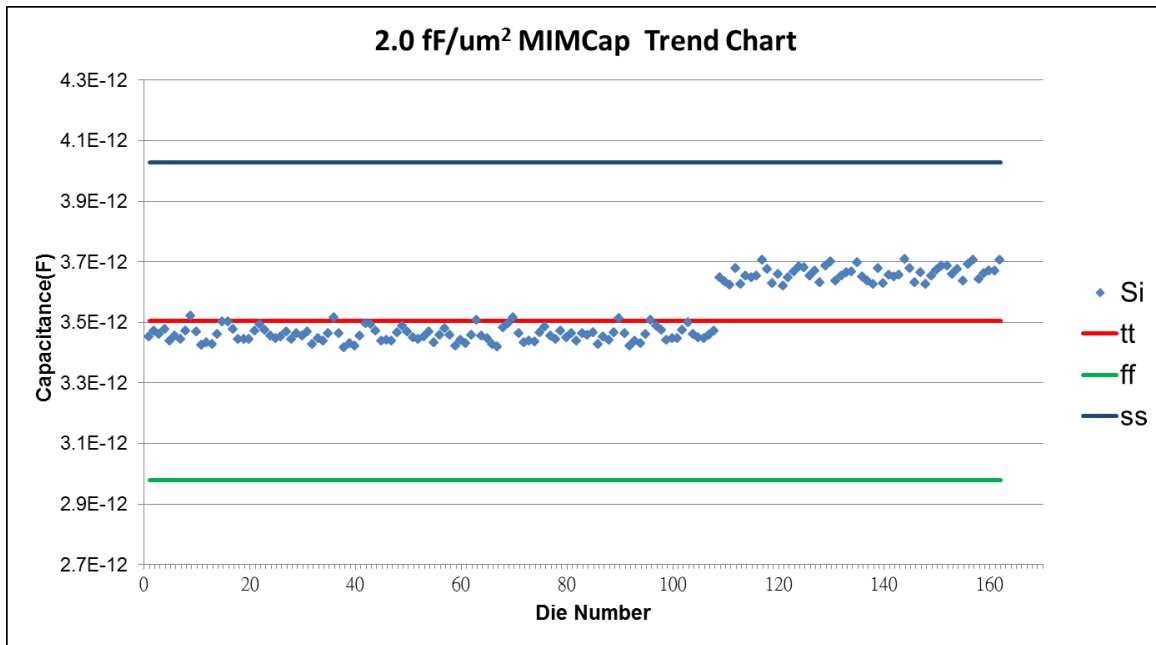


Figure 3-9 2.0 fF/um² MIMCAP_Trend Chart(Before shrink :W=48.6um , L=44.4um)

4 Appendix

4.1 HSPICE Warning Messages

Table 4-1 HSPICE Warning Message

NO.	Warning Message	Explanation
1.	None	None

4.2 ELDO Warning Messages

Table 4-2 ELDO Warning Message

NO.	Warning Message	Explanation
1.	None	None

4.3 SPECTRE Warning Messages

Table 4-3 SPECTRE Warning Message

NO.	Warning Message	Explanation
1.	None	None